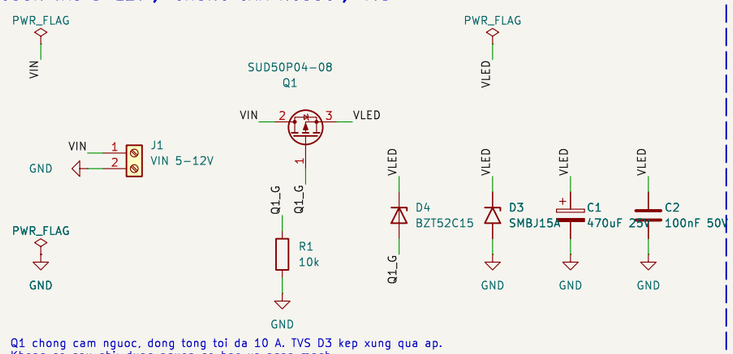
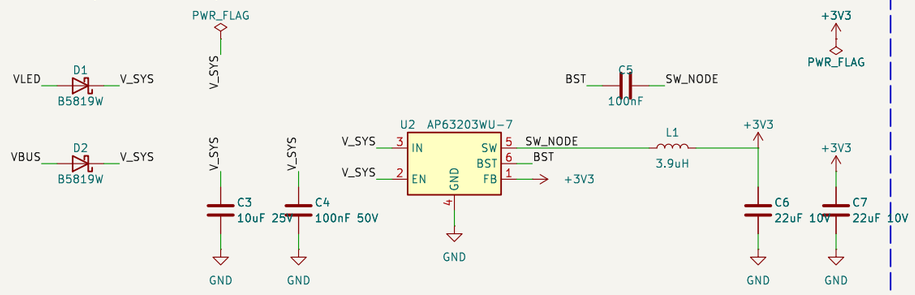
## NGUON VAO 5–12V / CHONG CAM NGUOC / TVS

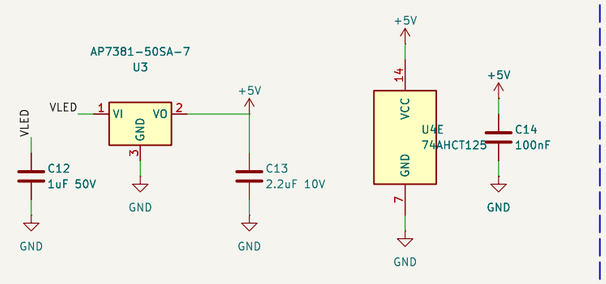



Q1 chong cam nguoc, dong tong toi da 10 A. TVS D3 kep xung qua ap.
Khong co cau chi: dung nguon co bao ve ngan mach.

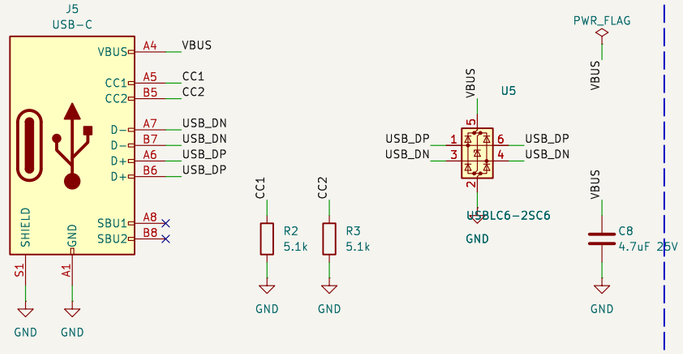
## NGUON 3.3V (buck AP63203)



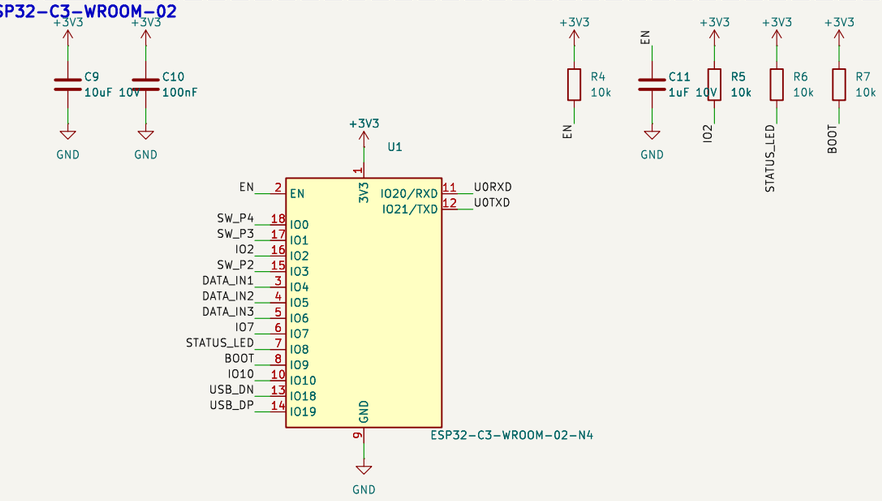

## 5V CHO IC DICH MUC (chi co khi co nguon LED)



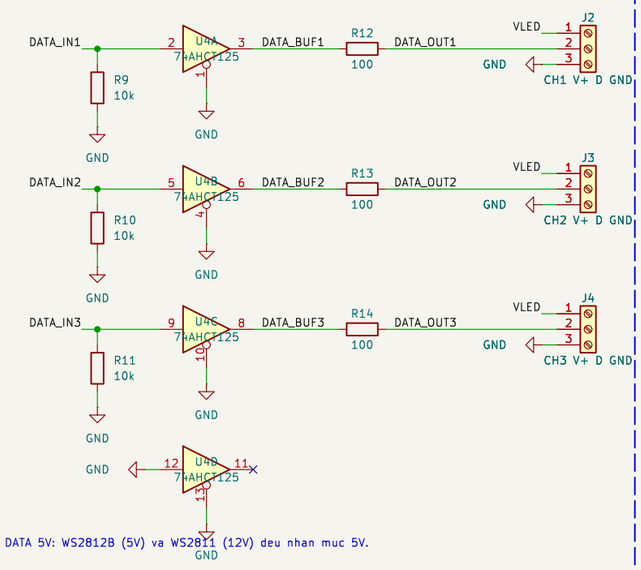

## USB-C: NAP CODE / DEBUG (USB Serial/JTAG)




## ESP32-C3-WROOM-02

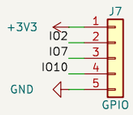



## DICH MUC 3.3V -> 5V + 3 KENH WS2812



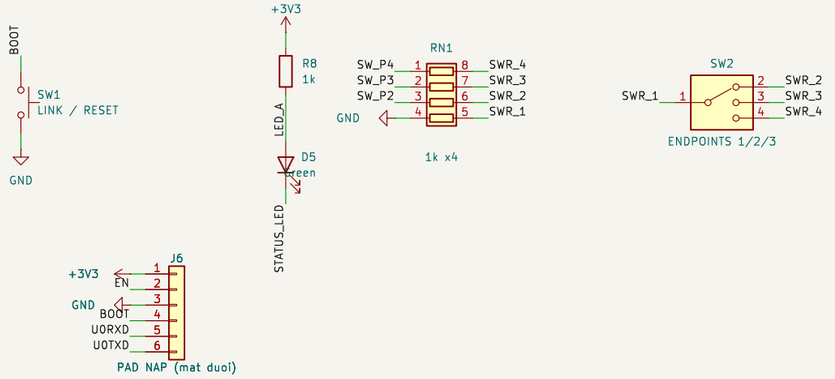

DATA 5V: WS2812B (5V) va WS2811 (12V) deu nhan muc 5V.

## MO RONG: 3 GPIO TU DO (J7)




IO2: chan strap, R5 10k keo len. Mach ngoai khong duoc keo xuong 0
khi ESP khoi dong. IO10: xung thap ~5 ns luc cap nguon.
IO7: chan GPIO thuong (MTDO).

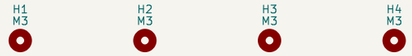
## NUT / LED / CONG TAC 3 NAC / PAD NAP




SW1 (GPIO9 = BOOT): bam nhanh = bat/tat hoac vao che do ket noi; giu >5 s = factory reset.
Giu SW1 trong luc cap nguon = vao che do nap ROM.

SW2: chan 1 -> GND (qua RN1), chan 2/3/4 -> IO3/IO1/IO0 (bat pull-up noi).
Firmware: chan nao doc 0 la dang noi voi chan 1; neu khong co, keo lan luot 1 chan xuong 0 va doc 2 chan con lai.
Nac = so thu tu chan lon nhat trong cap dang noi – 1 -> 1, 2 hoac 3 endpoint Matter.
Dung cho cong tac SP3T co chan chung la chan 1 hoac 2, ca kieu cau truot (1–2, 2–3, 3–4).

## LO BAT VIT M3




Generated by tools/build_sch.py from tools/design.py
Input 5–12V, 3x WS2812/WS2811 outputs, 1 button, 3-position endpoint switch

**Haxeko**

Sheet: /
File: c3_ws2812_3ch.kicad_sch

***Title: ESP32-C3 WS2812 3-channel Matter LED controller***

| Size: A3 | Date: 2026-11-09 | **Rev: 1.1** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 1/1 |